Categorizing Cyber-Attacks: Analysis of Industry, Event-Type, Economic and Criminological Factors

Problem Statement

Determine which industries are most affected by cyber-attacks, which types of attacks are most prevalent, and understand causes for certain industries being more vulnerable and attacked. Then, compare with other industries with lower cyber events to find unique causes.

Background Research

One reference is "A study of cyber attacks: In the healthcare sector" by Karuna Bhosale.

This reference highlights challenges including endpoint device management, security awareness issues, and inadequate protection of connected medical devices and hospital information systems

Limitation of prior work:
It does not identify which industry sector is most vulnerable to cyber attacks and why

Resolution of limitations:
My research addresses these limitations through a detailed data science analysis of cyber attack records to identify which industries are most impacted and the underlying causes

Hypothesis

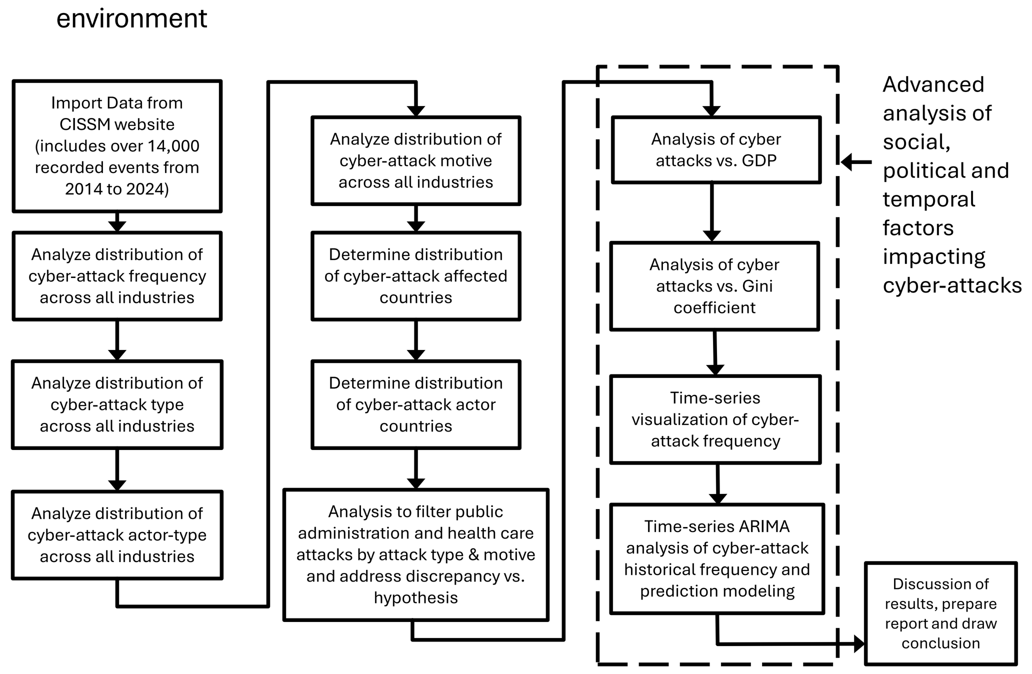
The healthcare industry has greatest number of cyber-attacks with the primary type being exploitative due to a large concentration of personal data, recurrent access to the data by numerous individuals, and potentially outdated systems.

Materials

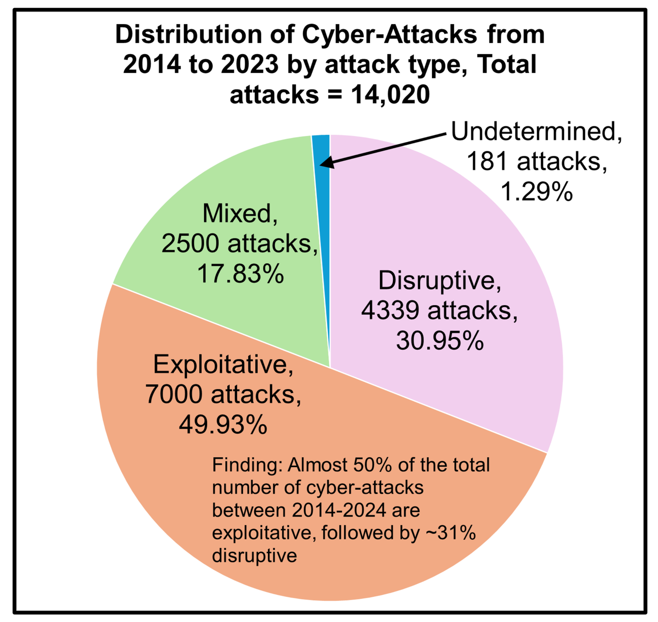
- This data science project uses data was sourced from Center for International and Security Studies at University of Maryland(CISSM).
  - The data, "Cyber Events Database" is a record of scraped publications of cyber-attack events occurring since 2014 to 2024 with over 14,000 recorded events.
  - The cyber events dataset used in this project was downloaded from this link:
    - https://cissm.umd.edu/cyber-events-database
- I also used data from The World Bank's Poverty & Inequality Indicators (PIP) which include various economic describing-variables per country including GDP, wealth gap, gini coefficient, etc.
  - I downloaded the PIP dataset that I used from this link:
    - https://pip.worldbank.org/poverty-calculator
- The cyber events are classified by date, actor type (criminal, hacktivist, nation-state, undetermined), actor, organization (which organization was impacted by the event), industry, motive (protest, sabotage, espionage, financial, undetermined), event type (disruptive, exploitative, mixed undetermined), event description, source url, target country, and actor country.
  - The data collects only publicly available information, using python to "scrape" data.
  - Then researchers from CISSM reviewed and categorized the data to ensure the source material information is reflected consistently in the data and the event identified fits their definition of a cyber event. The industries were categorized per the North American Industry Classification System.

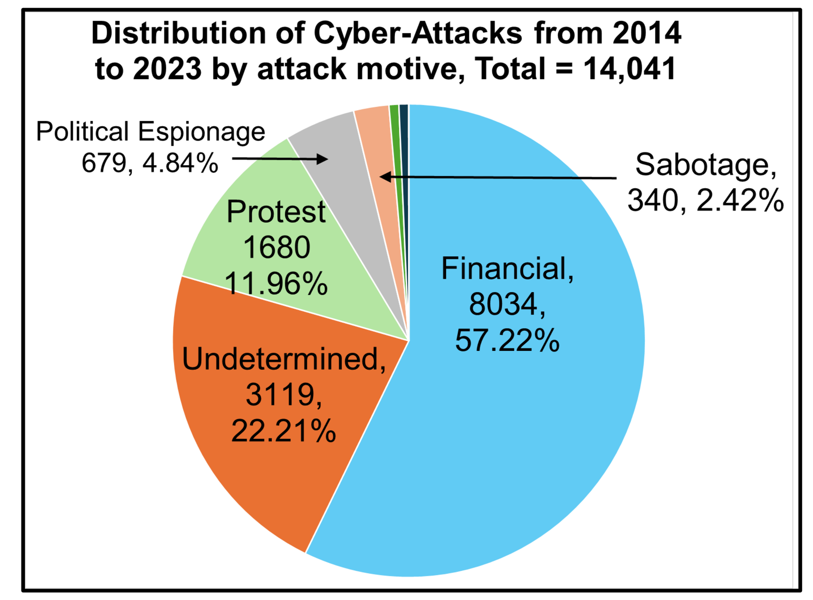
Procedures

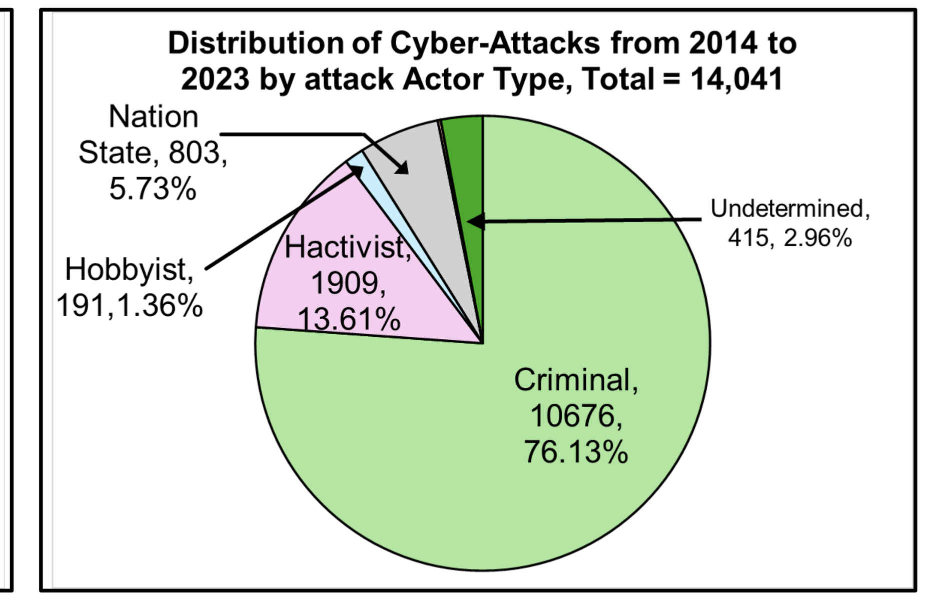
- The research followed the procedural steps in the block diagram flow chart below
- Implementation of these steps was done in Python in the Google colab environment

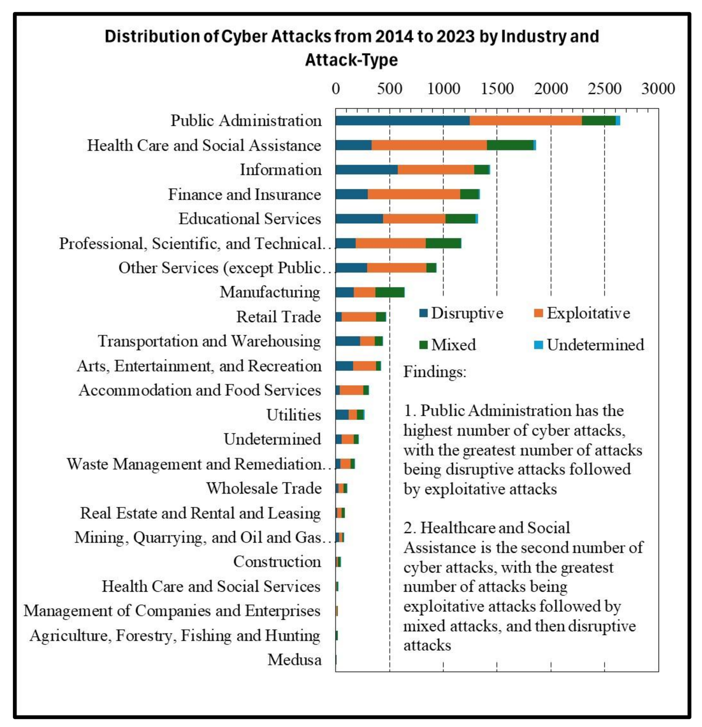


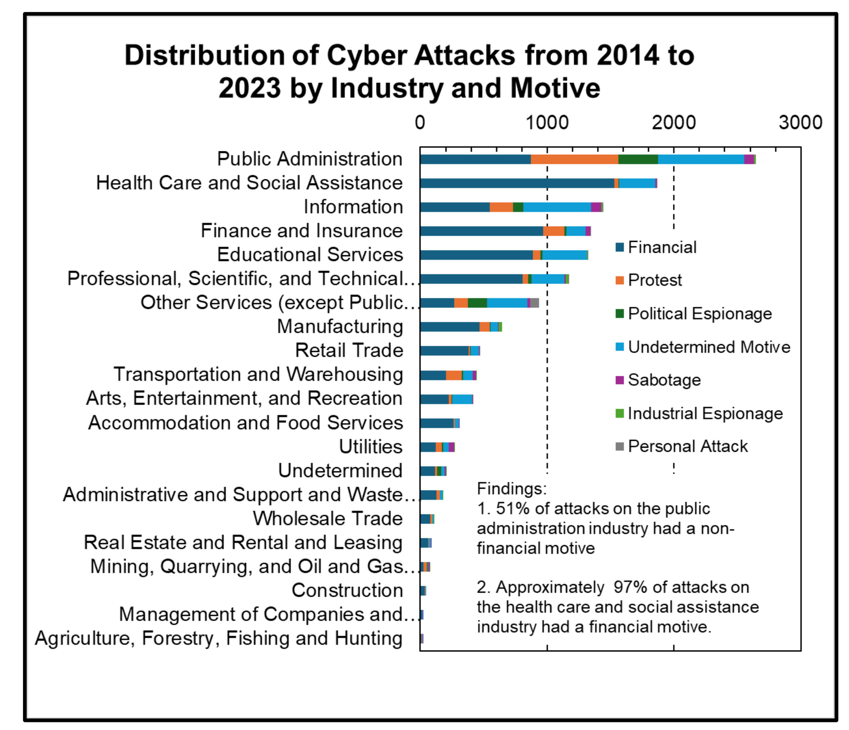
Results and Findings

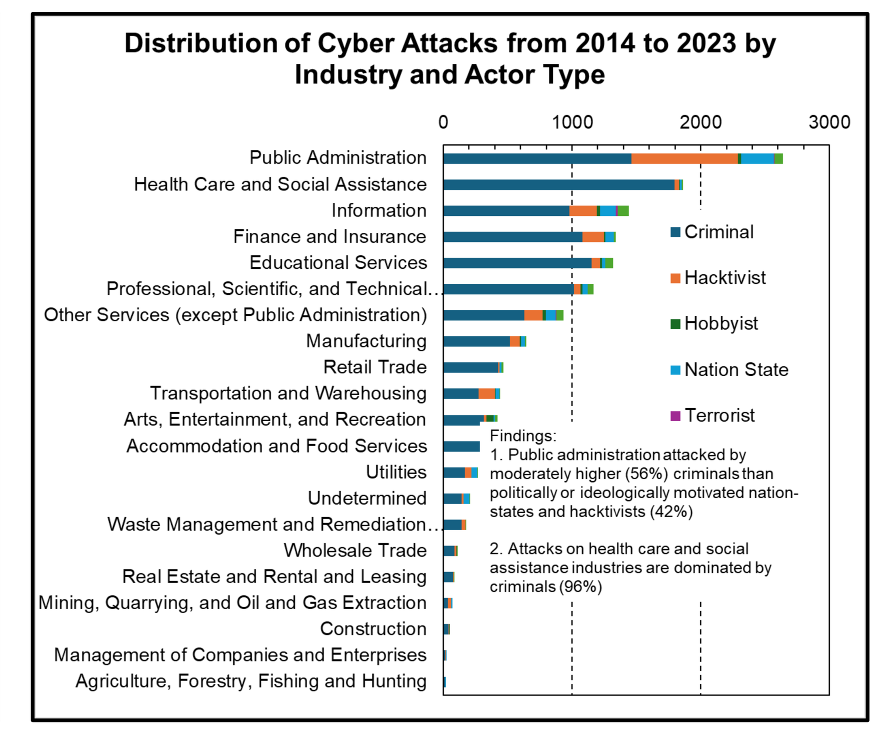


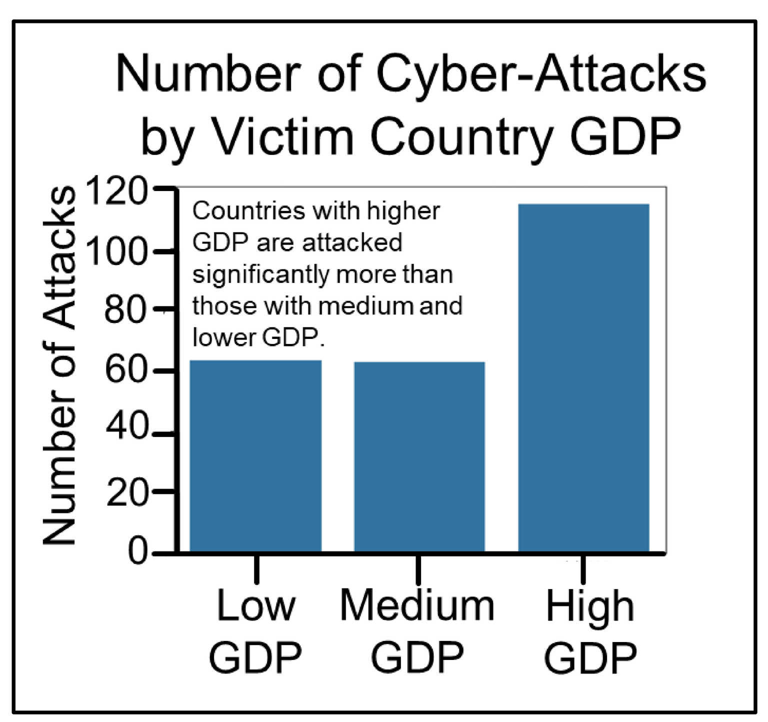


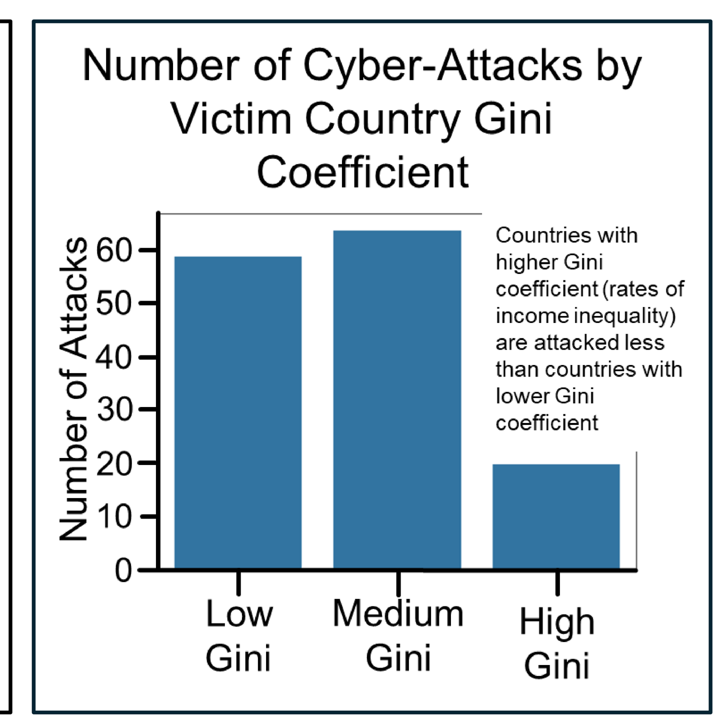


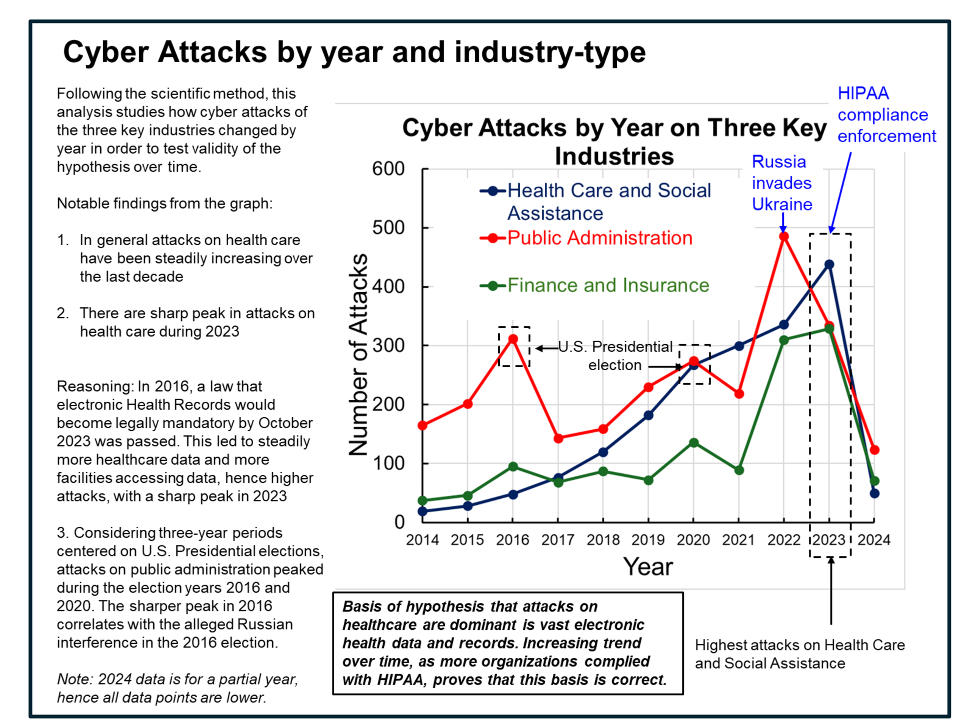


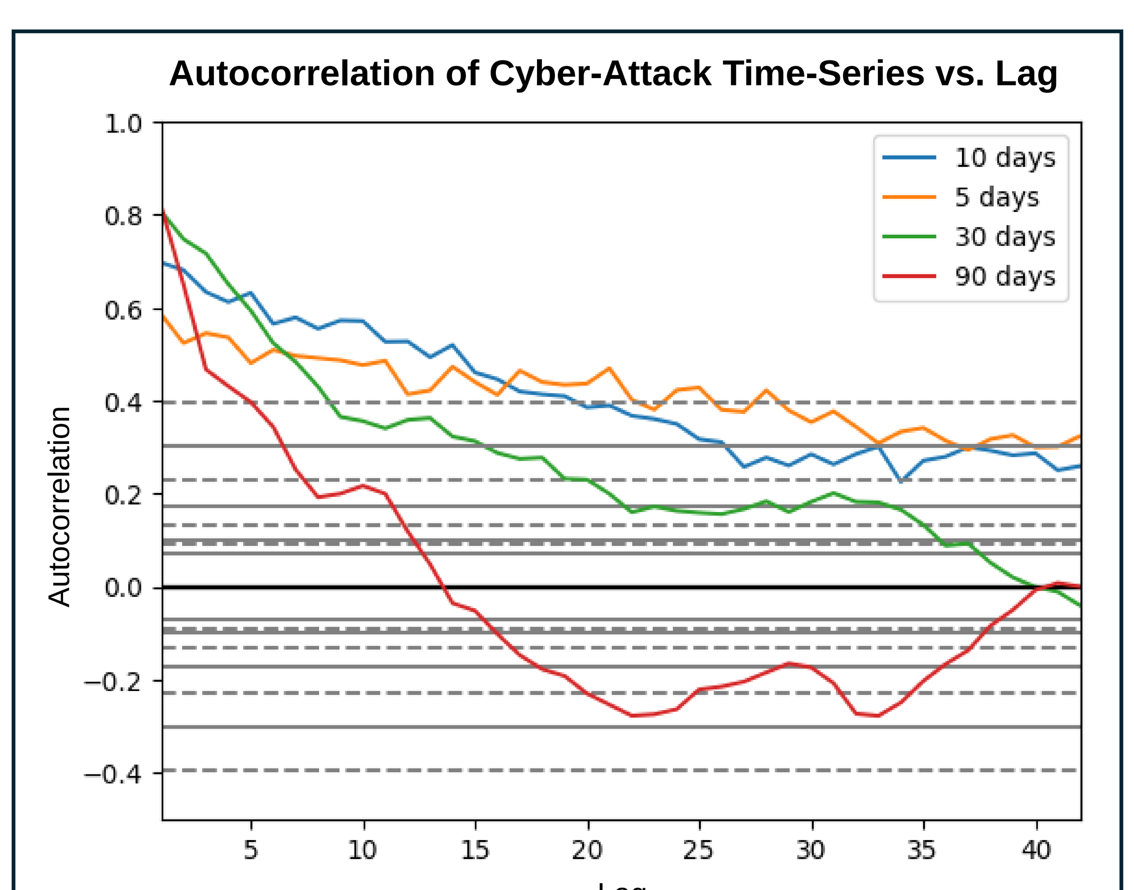


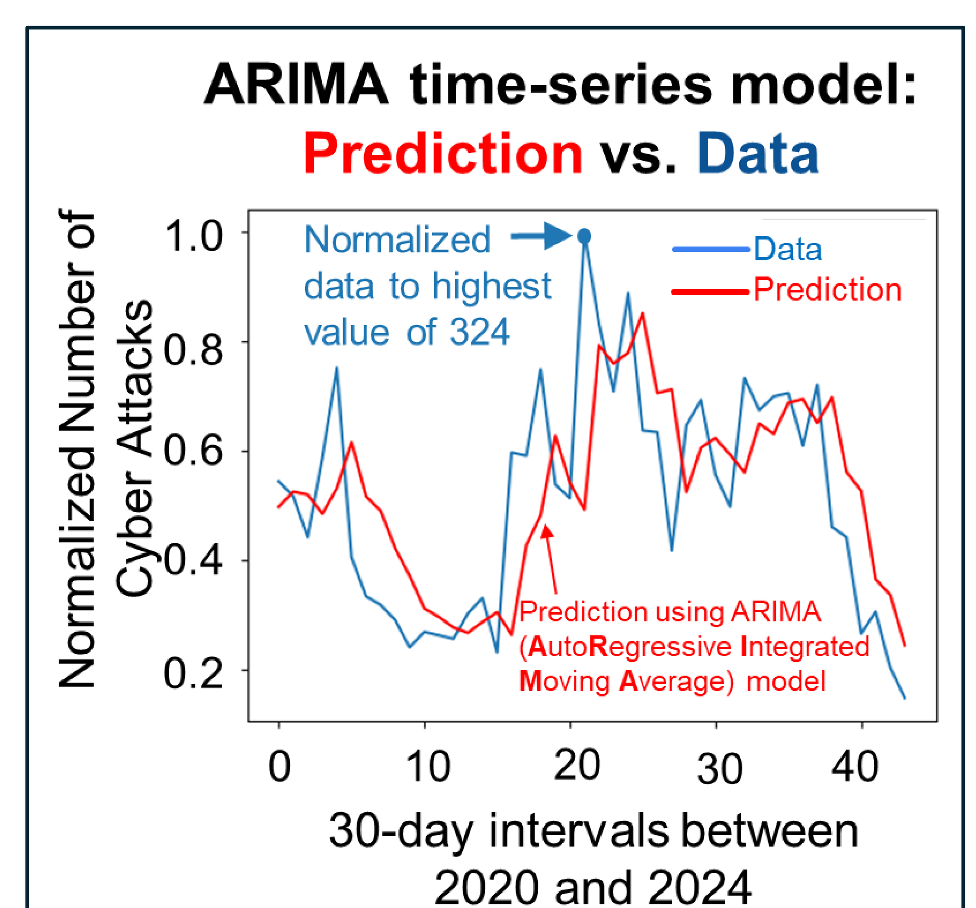












Discussion

- Consistent with the hypothesis, two out of three of the most recent full-years, i.e., 2021 and 2023, attacks on healthcare were highest. The increasing trend of attacks on healthcare and its correlation with the adoption of electronic health records agrees with the basis of the hypothesis.
- Considering the whole decade 2014 to 2024, The public administration industry has the highest number of cyber attacks, followed by health care and social assistance. This finding is close but not exactly consistent with the research hypothesis that health care would have the highest number of attacks. The work then proceeded with a detailed analysis of cyber attacks and their classification by attack type, motive and actors.
  - ➢ The findings indicate that attacks on public administration, and health care & social assistance differ in terms of these underlying factors: While public administration is found to have an approximately 20% higher number of disruptive attacks than exploitative attacks, the health care and social assistance industry is dominated by exploitative attacks which are approximately 3 times higher than disruptive attacks.
  - ➢ Additionally, in terms of actor type, public administration is attacked by a moderately higher (56%) criminals than politically or ideologically motivated nation-states and hacktivists (42%), while attacks on health care and social assistance are dominated by criminals (96%).
  - ➢ Then, from a motive perspective, 51% of attacks on public administration industry had a non-financial motive, whereas approximately 97% of attacks on the health care and social assistance industry had a financial motive.
  - ➢ Furthermore, the motive/actor profiles differ between the exploitative and disruptive attack types.
    - ✓ The motive of exploitative attacks is mostly financial (78%), while disruptive attacks have a slight majority of non-financial motives (52%).
    - ✓ Similarly, with actor profile, exploitative attacks have an overwhelming majority of non-political criminal actors (85%) while disruptive attacks also have a large prevalence of criminal actors, though less at 60% with a larger number of political actors.
- Considering cyber attacks only by non-political actors, i.e., not by nation-states or hacktivists, health care and social assistance is found to have the highest number of cyber attacks which supports the hypothesis of this research.
- Additionally, countries with high GDP and countries with greater equality (using Gini coefficient) are attacked more frequently
  - This is because such countries have more economic goods to capture and are more likely to be targets of political actors due to their greater influence across the world.
- Based on my ARIMA time-series model of the cyber attack data, it is clear that trends in cyber attacks are predictable.
  - This finding could be used to enhance security during periods of increased vulnerability.

Conclusion

- My hypothesis that healthcare would be victim to the greatest number of cyber-attacks was proved accurate in recent years. This is due to the only recent adoption of Electronic Health records as per legislation passed in 2016, coming into action by 2023 (wherin there was a clear spike in cyber attacks on the Healthare Industry)
- However, when considering only either exploitative attacks or those with financial motive, healthcare still has the greatest number of cyber-attacks over the whole 2014 – 2024 period
- This is due to the large concentration of personal data, recurrent access to the data by numerous individuals, and potentially outdated systems
- The only industry that surpassed health care in number of cyber was Public administration due to a large number of politically motivated disruptive attacks.
- This research identifies the industry with greatest number of cyber attacks and criminological and economic factors unlike prior studies found in my literature review:
- Politically motivated disruptive attacks resulted in Public administration having the greatest number of cyber attacks
- Financially motivated attacks resulted in many attacks on the healthcare industry
- Using GDP, countries with higher economic resources are attacked more frequently as attackers see a greater reward and that they are more influential on the global scene resulting in more enemies and politically-motivated attacks.
- Using Gini coefficient, countries with less inequality are attacked more frequently as those countries generally have greater overall prosperity to exploit.
- My time series analysis finds that there are predictable trends in cyber attacks that can help to enhance security at high-risk moments.
- Application-wise, my research can help to identify the where and why of cyber-attacks such that government policy and private cybersecurity work can focus on solving the problems where they exist
- Also, my work demonstrates the efficacy of using data science to recognize patterns in records of social science events.

Future Work

- This work predicts future attacks using the ARIMA model which predicts cyber-attacks based on historical data of prior attacks, but, does not consider underlying factors
- This work also includes data-science-based analysis and visualization of the underlying criminological factors (attack-type, motive, actor-type) and geopolitical factors (victim country GDP and Gini Coefficient)
- ***In the future, I would like to create an improved ARIMA or other time-series model which considers both the historical data and underlying factors, i.e., combines both of the above to be more accurate and help prevent cyber-attacks***